Programming Assignment #1

Assignment Description

Write an SIC assembler that reads an SIC assembly program, translates SIC statements into their machine code equivalents, and generates an object program.

Goals

1. Get familiar with C programming language.
2. Learn to use the I/O facilities and library functions provided by standard C.
3. Get experience with the system-level programming.
4. Get familiar with separate compilation, make utility, and C debugger.

Guidelines:

1. The score for a programming assignment includes demonstration in person (70%) and a report (30%) on regular due November 4th. You have to demonstrate your program to TA in person. For a two-member team, two programming assignments need to be demonstrated by different members. You will get 10% bonus if you succeed in demonstrating your program (without any obvious bugs) on October 28th noon, or earlier. On the other hand, you will be 10% penalty if you failed in demonstrating your program or handing in your report on December 9th noon, or earlier.
   1) (11/4th) Your score = X * 70% + Y * 30%
   2) (10/28th) Bonus demonstration = X * 70% * 1.1 + Y * 30%
   3) (12/9th) Late demonstration = X * 70% * 0.9 + Y * 30%
   4) (12/9th) Late demonstration + late report = X * 70% * 0.9 + Y * 30% * 0.9
2. You need to upload your report to www.elearn.ndhu.edu.tw
   1) Your report is suggested to include the following elements:
      i. A cover page
      ii. Brief description of the assignment problem
      iii. Highlights on how you write the program
      iv. Program listing
      v. Test run snapshots
      vi. Discussion
   2) Make your report in a pdf file, and upload to www.elearn.ndhu.edu.tw.
   3) The upload link will be suspended at the due time for a while for recording on-time hand-ins, and will be reopened for accepting late hand-ins. There will be a 10% penalty for lateness. (S = X * 70% + Y * 30% * 90%)

Programming Assignment #1 Addendum

Assembler Directives

START, END, BYTE, WORD, RESB, and RESW.

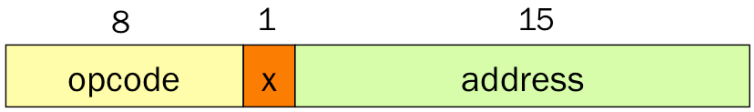
Format of SIC Assembly Program

1) Each line has 4 fields:
   i. (Optional) Label (up to 6 characters)
   ii. Operation
   iii. (Optional) Operands (no blank allowed)
   iv. (Optional) Comment
2) If the first character is a ".", the whole line is a comment.
3) There is no blank line in an SIC assembly program source file, except for the last line.

SIC OPCODE Table

Mnemonic	Opcode	Mnemonic	Opcode	Mnemonic	Opcode
ADD	18	AND	40	COMP	28
DIV	24	J	3C	JEQ	30
JGT	34	JLT	38	JSUB	48
LDA	00	LDCH	50	LDL	08
LDX	04	MUL	20	OR	44
RD	D8	RSUB	4C	STA	0C
STCH	54	STL	14	STSW	E8
STX	10	SUB	1C	TD	E0
TIX	2C	WD	DC		

SIC Instruction Format



Mode	Indication	Target Address
Direct	x = 0	TA = address
Indexed	x = 1	TA = address + (X)

Example Assembly Program

```
COPY    START   1000
FIRST   STL     RETADR
CLOOP   JSUB    RDREC
        LDA     LENGTH
        COMP    ZERO
        JEQ     ENDFIL
        JSUB    WRREC
        J       CLOOP
ENDFIL  LDA     EOF
        STA     BUFFER
        LDA     THREE
        STA     LENGTH
        JSUB    WRREC
        LDL     RETADR
        RSUB
EOF     BYTE    C'EOF'
THREE   WORD    3
ZERO    WORD    0
RETADR  RESW    1
LENGTH  RESW    1
BUFFER  RESB    4096
.
.       SUBROUTINE TO READ RECORD INTO BUFFER
.
RDREC   LDX     ZERO
        LDA     ZERO
RLOOP   TD      INPUT
        JEQ     RLOOP
        RD      INPUT
        COMP    ZERO
        JEQ     EXIT
        STCH    BUFFER,X
        TIX     MAXLEN
        JLT     RLOOP
EXIT    STX     LENGTH
        RSUB
INPUT   BYTE    X'F1'
MAXLEN  WORD    4096
.
.       SUBROUTINE TO WRITE RECORD FROM BUFFER
.
WRREC   LDX     ZERO
WLOOP   TD      OUTPUT
        JEQ     WLOOP
        LDCH    BUFFER,X
        WD      OUTPUT
        TIX     LENGTH
        JLT     WLOOP
        RSUB
OUTPUT  BYTE    X'05'
        END     FIRST
```

Types of Record in SIC Object Code

Header record:

Col. 1	H
Col. 2–7	Program name
Col. 8–13	Starting address of object program (hexadecimal)
Col. 14–19	Length of object program in bytes (hexadecimal)

Text record:

Col. 1	T
Col. 2–7	Starting address for object code in this record(hexadecimal)
Col. 8–9	Length of object code in this record in bytes (hexadecimal)
Col. 10–69	Object code, represented in hexadecimal (2 columns per byte of object code)

End record:

Col. 1	E
Col. 2–7	Address of first executable instruction in object program (hexadecimal)

Example Object Program

```
HCOPY  001000001107A
T001000 1E 141033482039001036281030301015482061 3C1003 00102A 0C1039 00102D
T00101E 15 0C1036482061 0810334C0000 454F46 000003 000000
T002039 1E 041030001030 E0205D30203F D8205D 281030 302057 549039 2C205E 38203F
T002057 1C 1010364C0000 F1 001000 041030 E02079 302064 509039 DC2079 2C1036
T002073 07 3820644C000005
E001000
```